

Vector

One advantage of using a vector is that it is simple to build and easy to understand. The data is stored in a single collection, which makes loading and displaying course information straightforward. Vectors also use memory efficiently because they do not require additional pointers or hashing functions. A disadvantage is that searching for a specific course can be slow because the program may need to check every course until it finds a match. Another drawback is that the courses must be sorted before they can be displayed in alphanumeric order, which adds extra processing time.

Hash Table

A hash table is useful because it allows the program to find a course very quickly using the course number as a key. This makes searching much faster than a vector when working with a larger amount of data. Another advantage is that inserting course records into the table is efficient. One disadvantage is that the data is not stored in sorted order, so additional steps are required before printing the course list alphabetically. Hash tables can also experience collisions, which may reduce performance if multiple courses are assigned to the same location in the table.

Binary Search Tree

A binary search tree provides a good balance between searching and sorting. Since the data is stored in order, the program can print the course list alphabetically by performing an in-order traversal of the tree. Searching for an individual course is also efficient because the program only needs to follow a portion of the tree rather than checking every course. The main disadvantage is that binary search trees are more complex to implement than vectors or hash tables. In addition, performance can decrease if the tree becomes unbalanced, causing searches and insertions to take longer than expected.

Conclusion

After comparing all three data structures, I would choose the binary search tree for the final project. The vector is easy to use but requires more time when searching for courses. The hash table provides the fastest search performance, but the data is not automatically stored in order. The binary search tree offers the best combination of searching and sorting because courses can be located efficiently while still being displayed in alphanumeric order without performing an additional sort. Since the advising program needs both quick lookups and an ordered course list, the binary search tree is the best overall choice.

Data Structure	Load Data	Search Course	Print Sorted List	Memory Usage
Vector	$O(n)$	$O(n)$	$O(n \log n)$	$O(n)$
Hash Table	$O(n)$	$O(1)$ average	$O(n \log n)$	$O(n)$
Binary Search Tree	$O(n \log n)$	$O(\log n)$	$O(n)$	$O(n)$